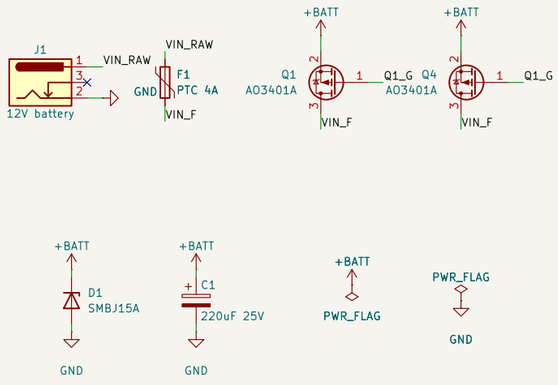
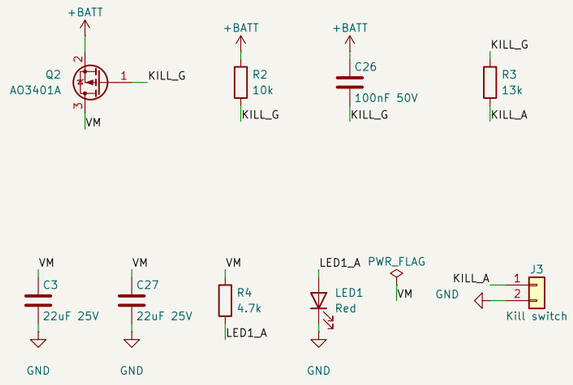
Battery in: fuse and reverse-plug protection



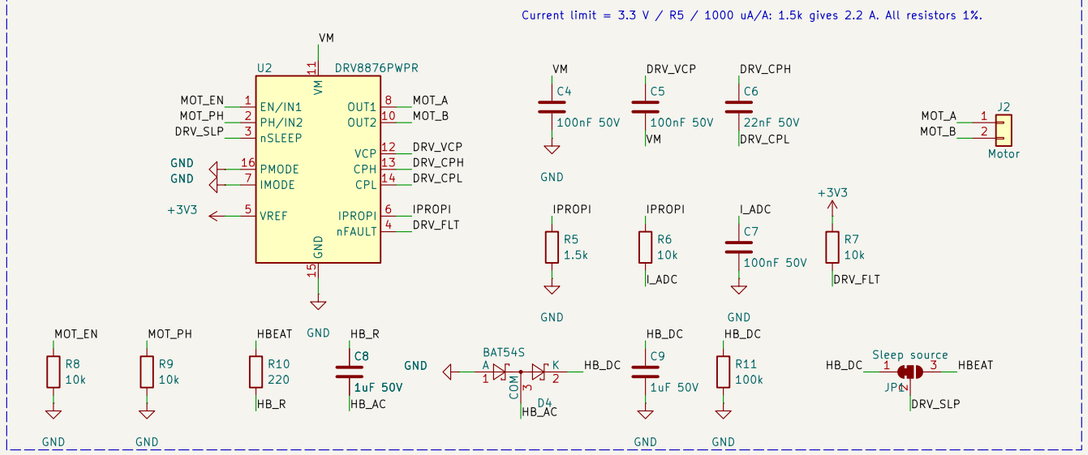
3S lithium-ion pack, 9.0 to 12.6 V, on its barrel plug (centre positive). Q1 and Q4 stay off if the plug is wired the other way round. R34 and R1 hold their gates at a third of the battery.

Motor supply and kill switch

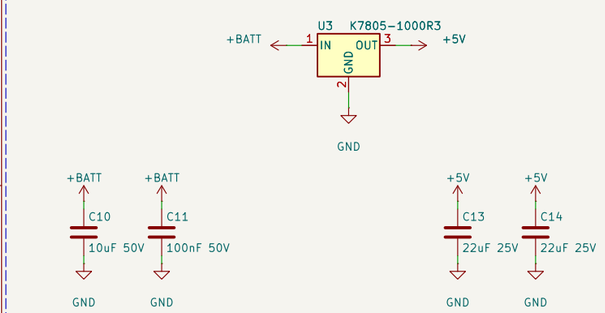


Switch closed (or a jumper on J3): motor supply on, Q2's gate at 0.43 of the battery. Open, unplugged or a broken wire: off within about 2 ms. C26 keeps Q2 off at plug-in.

Motor driver, current limit and watchdog

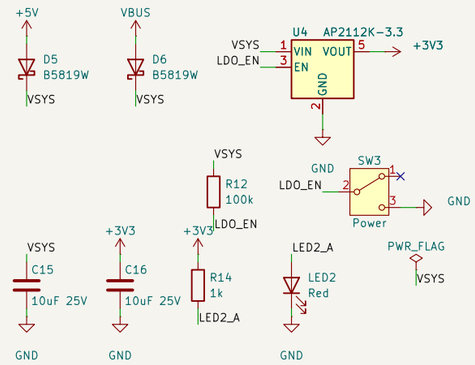


12 V to 5 V: ready-made module

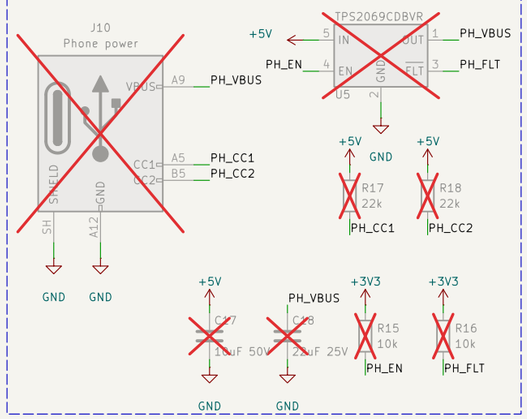


A sealed switching module (YLPTEC K7805-1000R3: 8-36 V in, 5 V 1 A out, pins 1 in, 2 ground, 3 out). The converter's layout is inside it, so the board's layout cannot spoil it.

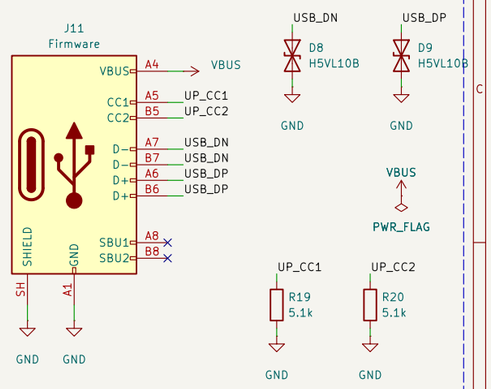
3.3 V



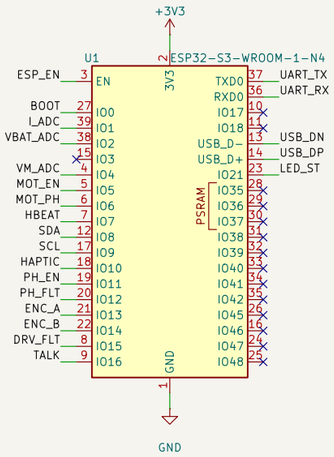
5 V out for the phone (not fitted)

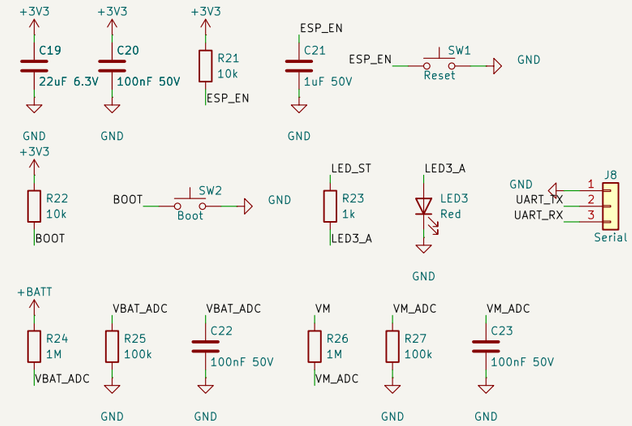


USB-C for firmware



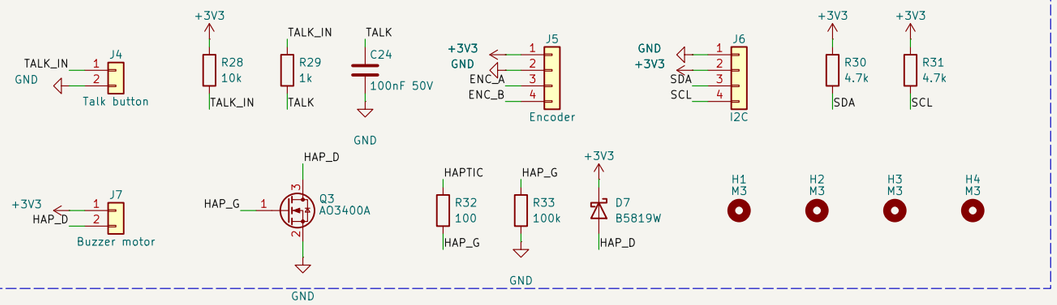
Processor: ESP32-S3





IO3, IO45, IO46 are boot-mode pins: left open. IO35 to IO37 are left open so a module with more memory also fits. Dividers 1M/100k: 2.2 V at the 24.4 V clamp, and only microamps into the processor when it is off.

Talk button, encoder, I2C, handle buzzer



Drawn by make_board.py. Not built or tested.

Sheet: /
File: cane-board.kicad_sch

Title: SightSense cane board

Size: A3	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/1